Setting groups as programs

Calculator settings on the C47 and R47 can be programmed and kept as reusable quick sets. The specific settings shown here are entirely user-selectable and mainly reflect the author's own preferences. They are not recommendations and are included only to illustrate the method used in this app note.

This note originates from a forum post dated 2024-08-02 and follows that post closely. Only the settings discussed there are described and only those settings appear in the sample program. The purpose is not to review all available settings, but to introduce a small set that many engineers tend to change and to show how such settings can be grouped, saved, and reused.

All settings are programmed in PRGM and managed using WRITEP, READP, and XPORTP. Programs are saved via I/O → WRITEP and loaded via I/O → READP. A human-readable version can be archived using I/O → XPORTP. The settings shown are typical rather than authoritative and should be adapted freely by the user.

Additional settings

A second list of settings is provided for reference only. These were not part of the original forum post and are not included in the sample program. They are noted simply as related or interesting areas for further investigation when building personal quick sets.

Sample program

The program SETT.p47 has been part of the distribution PROGRAM sample files since original discussion on the forum. Refer to the downloadable distro sample programs.

Sample program SETT.p47:

```
C47 Program file export: Export format version 3, C47 program version 1.
0000:  Prgm #43/43: 156 bytes / 29 steps

0001:  LBL ‘SETT’
0002:    REM ‘Setting example program, for Appnote15, 2024-08-01’
0003:    REM ‘Updated 2026-01-25’
0004:    S.EUR
0005:    RWDOT ·
0006:    MULT·
0007:    CPXⅈⅉ
0008:    CF ‘CPXmul’
0009:    CF ‘CONVHP’
0010:    FDIGS 31
0011:    CF ‘FRCYC’
0012:    SF ‘IRFRAC’
0013:    SF ‘PROPFR’
0014:    SF ‘DENANY’
0015:    CF ‘DENFIX’
0016:    DMX 12
0017:    SIG 03
0018:    ENGOVR
0019:    HIDE 31
0020:    RNG 6143
0021:    SF ‘dℤ.0’
0022:    CF ‘dLrgLI’
0023:    CF ‘RPHP’
0024:    SF ‘eRPN’
0025:    SSIZE8
0026:    SF ‘CPXRES’
0027:    SF ‘SPCRES’
0028:    CF ‘1024n’
0029:  .END.
```

The listed typical settings for operational engineering essentials, differing from the C47 / R47 defaults are discussed below:

Manual	Setting	Program	Description
DISP p3, F2	EUROPE	SETEUR	Set typical European settings and change preferred details below. Various bottled configs to setup regional settings.
DISP gF2	RADIX	WDOT•	Choose Radix to be "wide dot". Also change IPART and FPART to customize number formats.
DISP p2 fF6	Set/clear MULT·	MULT·	Choose the multiplier to be · Multiplier is used for ***i***×2 and 2×10^2 Alternatively use MULT×
DISP p2 fF2	Set/Clear CPXj	CPXj	Choose complex operator to be *j* Math way to display i EE way to display *j*
DISP p2 fF3	Set/Clear CPXmul	CF SYS.FL CPXmul (prgm in SYS.FL)	Choice: SF:1.+***i***×2.; CF:1.+2.***i*** Set for scientific way of writing complex numbers, 1.0+***i***×2.0 instead of default 1.0+2.0***i***
MODE/PREF CFLG	Set/Clear $CONV_{HP}$	CF SYS.FL $CONV_{HP}$ (prgm in SYS.FL)	Clear to have the CONV menu button to be the destination unit, example: Default is set: Clear: [km→ | ←mi.]: 1 [km→] means "convert 1 km to miles". Set: [km ←|→ mi.]: (HP = default way) 1 [km←] means "convert 1 mile to km".
DISP p1 gF4	FDIGS 31	FDIGS 31	Set fraction tolerance to be 10^-31 Set to show fractions without ≈, < and > if accurate within ±1E-31.
DISP p1 gF5	Clear FRCYC	CF SYS.FL FRCYC (prgm in SYS.FL)	Clear to not use the full cycle on [a b/c]. SF:cycle, CF:on/off. Set to change the [a b/c] button to cycle through all fraction combinations.
DISP p1 fF2	Set IRFRAC	SF SYS.FL IRFRAC (prgm in SYS.FL)	Set to enable irrational display Set to see 2π etc. instead of 6.28… Statusbar: "·𝕀"
DISP p1 fF3	Set PROPFR	SF SYS.FL PROPFR (prgm in SYS.FL)	Set to allow mixed fractions Clear for $5e/2$ instead of 2½ e. Statusbar: "b/" or "ab/".
DISP p1 fF5	Set DENANY	SF SYS.FL DENANY (prgm in SYS.FL)	Fractions in full denominator range. Statusbar: "b/200" with no "x", or "f" DENANY DENFIX Statusbar

DISP p1 fF6	Clear DENFIX	CF SYS.FL DENFIX (prgm in SYS.FL)	clear (def) clear (def) × or · clear set f set clear " " (blank) Not active in IRFRAC mode. × or · denominator fixed but fractions simplified, for DMX=12: 1/12, 1/6, 1/4, 1/3, 5/12, 1/2, 7/12, 2/3, 3/4... f denominator fixed, for DMX=12: 1/12, 2/12, 3/12, 4/12, 5/12… " " any fraction with denominator 1-DMX
DISP p1 fF4	DMX 12	DMX 12	Choose 12 for ½ ¼ ⅜ etc. Popular settings are 120 for $^{355}/_{113}$; 32 for inches etc., 180, 999, 1200. Fraction denominator maximum. Statusbar: "/max" or "/200".
DISP p1 F5	Set SIG 3	SIG 3	Set to SIG 3 significant digits. Pick display mode and digits: Measurements and gauge calcs: SIG 3 Electronics: UNIT 3 General engineering: ENG 4 General: ALL 3 FIX..ALL, [.d] etc. will override IRFRAC. Switch back on with [a b/c].
DISP p2 gF2	Set ENGOVR	ENGOVR (alt. prgm in SYS.FL as ALLENG)	Set to ENG for over range, choose ENG/SCI: The ALL parameter specifies where to 'jump', eg: ALL 4 will display 0.001 and 0.0001 ALL 3 will display 0.001 and 100.×10-6
DISP p2 gF6	HIDE 31	HIDE 31	≈0 for numbers < 1E-31. Mute exponents shown for minute numbers <|1E-31|. Example: 1×10-32 displays as ≈0 10×10-32 displays as 1×10-31
DISP p2 gF5	RNG = default 6145	RNG 6143	Positive and negative exponent limit Default exponent range, ±6145 may be too high. Change to 99 to be closer to conventional calculators.
DISP p2 F2	Set d$\mathbb{Z}$.0	SF SYS.FL d$\mathbb{Z}$.0 (prgm in SYS.FL)	Set for integers and reals displayed in the same FIX…ALL notation. Integers marked with $\mathbb{Z}$: indicating no decimals and 1000 digits
DISP p2 F1	Clear LRG_LI	CF SYS.FL LRG_LI (prgm in SYS.FL)	Integer large font. Clear to see up to 42 digits of integer digits in small font

C47 MODE p1 fF5 R47 PREF p1 F5	Clear RP_{HP}	CF SYS.FL RP_{HP} (prgm in SYS.FL)	Clear RP_{HP} to have →R →P X/Y entry sequence same as the COMPLEX sequence. Set RP_{HP} to have →R →P entry as on HP15C et al.
C47 MODE p1 fF4 R47 PREF p1 F4	eRPN default is set	SF SYS.FL eRPN (prgm in SYS.FL)	No ENTER duplicate. Clear for classic RPN (duplicate entry)
C47 MODE p2 F2 R47 PREF p1 F4	SSIZE8 default 8	SSIZE8	Stack size 8 (default 8)
C47 MODE p1 F3 R47 PREF p2 F3	CPXRES default is set	SF SYS.FL CPXRES (prgm in SYS.FL)	Allow Complex results without errors (default)
C47 MODE p1 F4 R47 PREF p2 F4	SPCRES default is set	SF SYS.FL SPCRES (prgm in SYS.FL)	Allow ∞ results without errors (default)
C47 HOME Prefix gF6 R47 PFX gF6	1024^n = default is clear	CF SYS.FL 1024^n (prgm in SYS.FL)	Clear 'data' prefixes. Decimal/binary setting for UNIT disp mode: When set, 1Mi means 1024^2 = 2^10 in contrast when clear, 1M means 1000^2 = 10^6. Set to change UNIT to binary prefixes, ranging from kibi ($2^{10\times1}$) to pebi ($2^{10\times5}$)

Other selected interesting settings/buttons, not in the SETT.p47 program

[a b/c]	--	--	Re-activate IRFRAC. When UNIT is set after IRFRAC, clearly UNIT de-activates IRFRAC. Clear the fractions with [.d] and set with [a b/c]
MODE/PREF CFLG	$BASE_{HP}$ def set	SF $BASE_{HP}$ (prgm in SYS.FL)	Clear to have HEX/DEC/OCT/BIN/# control the X-register only, for when you need X in a different base than other stack levels Y-T.
DISP F4	dSI def 2	dSI 1	Change to 0 to have no special BASE screen, 1 to have one top line only, and 3 to have 3 lines of BASE screen.
MODE/PREF CFLG	PFX.All def clear	SF PFX.All	Set to expand the UNIT range to quecto (10^{-30}) - quetta (10^{30}) where the default allows SI prefixes from femto (10^{-15}) - peta (10^{15}).
C47 MODE F4 R47 PREF F3	SDIGS def 34	SDIGS 16	Significant digits. Change 0-34 to increase solving and integrating speed by reducing accuracy. This influences all calculations, not only the solver. Remember to set it back.
C47 MODE p2 R47 PREF p2	iLI/RL def iR iC iLI	iREAL	Set by default to iLI/RE input mode. Integers are input if no decimals are found, and reals if decimals found. iR could be selected to force REAL only input.

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